🎯 Visual Tracking and Grasping Dynamic Color Blocks

1. Course Overview

The function implemented in this lesson can be simply understood as: **robotic arm "eyes" fixate on the specified color block, automatically follow movement, and approach to grasp when aligned**.

By extracting the target block's centroid in the HSV color space, using PID closed-loop control of the robotic arm's joint 1 base servo and joints 2/3/4 pitching, the camera always faces the block directly. After the block centroid is continuously stable in the image center, the chassis automatically advances to the grasping distance (approximately 230~240mm), and the robotic arm executes IK descent to complete grasping.

> **Note**
>
> **The tutorial recommends using the accompanying 4×4cm colored wooden blocks. The larger size helps improve tracking stability.**

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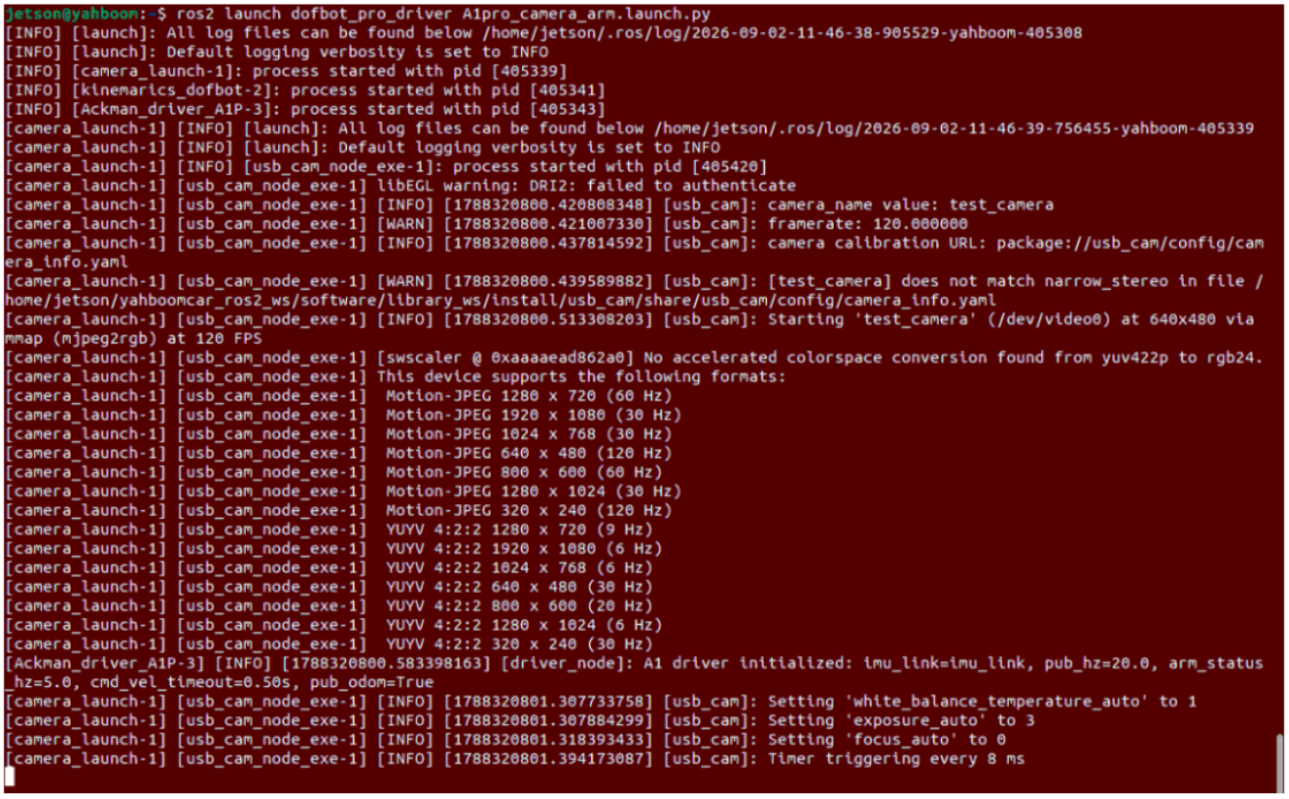
2. Preparation and Running

This case requires opening 3 terminals to start the system sequentially:

2.1 Start Camera Driver and Kinematics Solver Environment

Run in **Terminal 1**:

```
ros2 launch dofbot_pro_driver A1pro_camera_arm.launch.py
```

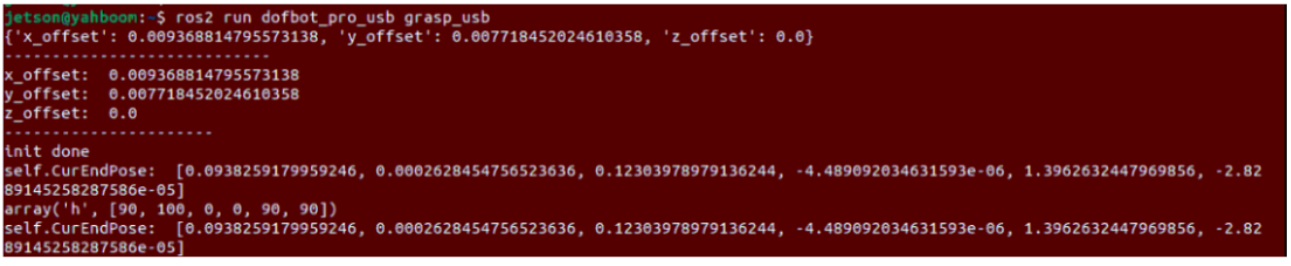


2.2 Start Robotic Arm Desktop Grasping Action Execution Node

Run in **Terminal 2**:

```
# Depth camera version (Nuwa)
ros2 run dofbot_pro_driver grasp_desktop

# Or USB camera version
ros2 run dofbot_pro_usb grasp_usb
```



2.3 Start Color Block Tracking Node

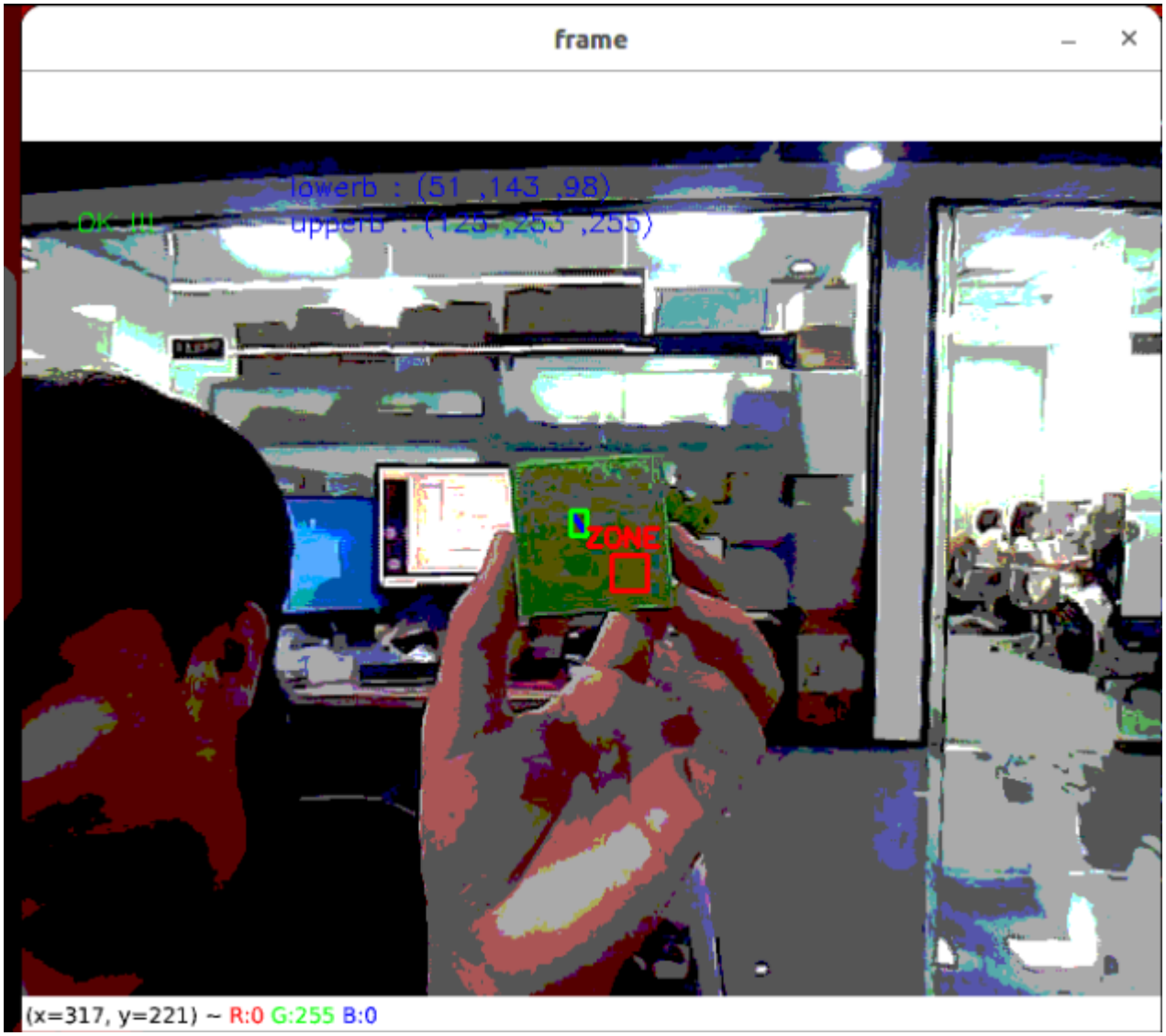
Run in **Terminal 3**:

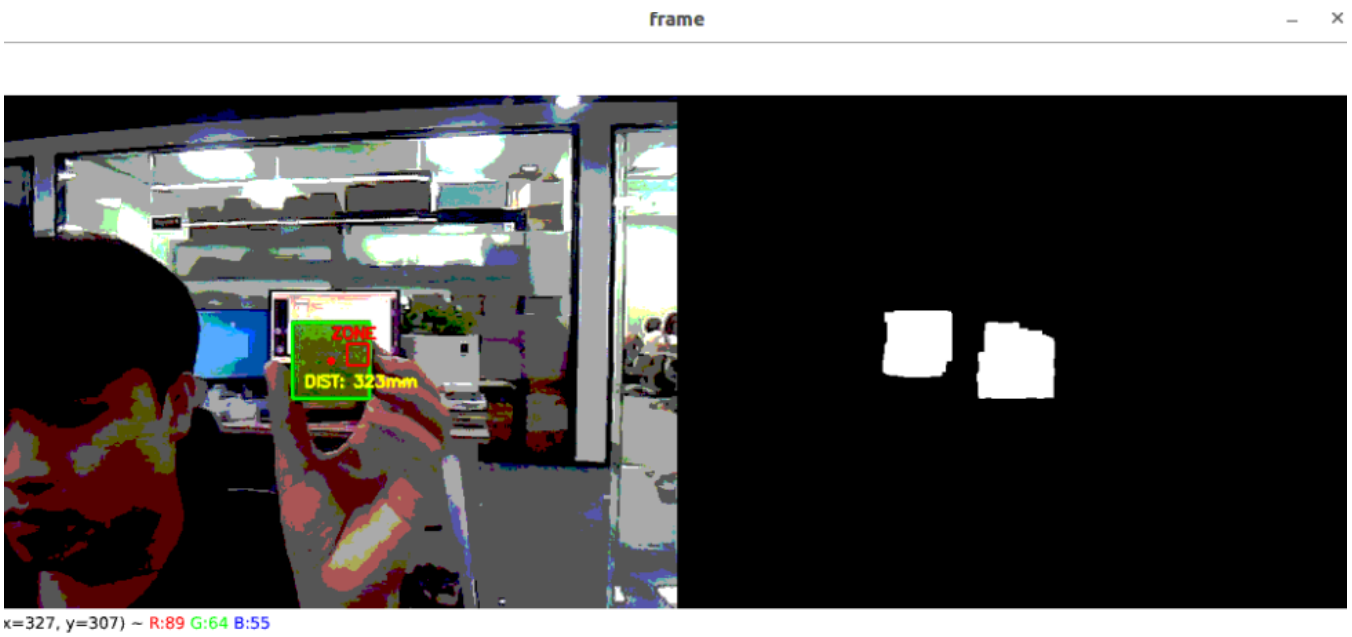
```
# Depth camera version (Nuwa)
ros2 run dofbot_pro_driver color_follow_2D

# Or USB camera version
ros2 run dofbot_pro_usb color_follow_usb
```

```
jetson@yahboom:~$ ros2 run dofbot_pro_usb color_follow_usb
----------------------
[INFO] [1788335893.297666263] [ColorRecognize_node]: Loaded Offset -> x: 0.009368814795
573138, y: 0.007718452024610358, z: 0.0
[INFO] [1788335893.299933561] [ColorRecognize_node]: object_size: 0.04 m, depth_scale:
0.936504
[INFO] [1788335893.318143692] [ColorRecognize_node]: Color tracking and grasping node (
USB monocular edition) started successfully!
```

Mouse click to select color:





3. Operation Demo and Interaction Flow

1. **Select Target Color**:
    - With the image window active, press the **C key** to enter color selection mode;
    - Use the mouse to drag and select the block. Resample HSV thresholds under the current lighting environment and automatically save to `custom_colorHSV.text`. The system immediately locks onto that color target and enters servo tracking;
2. **Handheld Material Guidance Tracking**:
    - Hold the 4×4cm block and slowly move it within the field of view in front of the robotic arm;
    - The robotic arm will automatically perform gimbal PID servo rotation based on the block centroid offset, keeping the camera crosshair aligned with the block;
3. **Center Locking and Chassis Approach**:
    - After the block centroid stably enters the center target box, the gimbal pose locks, and the chassis automatically and smoothly drives forward toward the block;
4. **Automatic Grasping Trigger**:
    - After reaching optimal grasping distance, chassis stops, buzzer sounds, robotic arm automatically computes the gripper alignment angle and descends to grasp;
    - After grasping, the robotic arm lifts the material and resets, resetting status flags to prepare for the next tracking cycle.

> **Tip**
>
> When moving the block, please **move slowly and smoothly**. Too fast movement can cause motion blur in the image and lead to PID tracking overshoot.

4. Core Source Code Analysis

The system is completed by "color block tracking decision" and "IK execution" two nodes working in coordination:

Camera Type	Color Block Tracking Decision Node	Grasping Action Execution Node
USB Monocular Camera	`dofbot_pro_usb/color_follow_usb.py`	`dofbot_pro_usb/grasp_usb.py`
3D Depth Camera	`dofbot_pro_driver/color_follow_2D.py`	`dofbot_pro_driver/grasp_desktop.py`

Regardless of which camera is mounted, the program underlying is designed as a **four-stage state machine**:

```
Phase A: Gimbal PID Tracking -> Phase B: Continuous Center Locking -> Phase C: Chassis Active Approach -> Phase D: Gripper Adaptive Grasping
```

The main technical implementation comparison between the two:

Comparison Dimension	USB Monocular Camera Solution	3D Depth Camera Solution (Nuwa-HP60C)
Alignment Center Reference	Image optical center `TARGET_X = 341, TARGET_Y = 242`	Image standard center `TARGET_X = 320, TARGET_Y = 240`
Lock Transition Mechanism	Switch to approach immediately when block centroid enters center tolerance box	Continuous 10 frames lock (`ALIGN_COUNT_THRESHOLD = 10`) before confirming approach, stronger anti-noise capability
Ranging and Grasping Distance	Pinhole calculated distance, approach to approximately 230mm	Depth map matrix directly reads true millimeter depth, approach to approximately 240mm
Spatial Compensation Dimension	Superimpose X, Y 2D spatial calibration deviation	Synchronously superimpose X, Y, Z 3D omnidirectional compensation from `offset_value.yaml`

4.1 USB Monocular Camera Core Source Code Analysis (`color_follow_usb.py`)

1. Gimbal PID Dynamic Following and Slope Limiting

Calculate pixel deviation between block centroid and target optical center, drive joints through PID with slope limiter (`max_step_change = 0.03`) to prevent jitter:

```
# color_follow_usb.py -- XY_track gimbal tracking
raw_angular_x = self.angular_x_PID.compute(center_x, TARGET_X)
raw_angular_y = self.angular_y_PID.compute(center_y, TARGET_Y)

# Slope limiter smoothing filter
diff_x = raw_angular_x - self.last_angular_x
angular_x = self.last_angular_x + max(-self.max_step_change, min(self.max_step_change,
diff_x))

target_j1 = 90  + (10 * angular_x * -90.0)   # Base rotation
target_j2 = 150 + (angular_y * -150.0)        # Pitch coordination
target_j3 = 12  + (angular_y * -60.0)
target_j4 = 20  + (angular_y * -100.0)
```

2. Chassis Approach and Descent Grasping

After centroid alignment and gimbal lock, chassis uses forward/backward PID to approach 230mm, combined with block bounding box corner vectors `compute_joint5(vx, vy)` to calculate gripper alignment angle and trigger grasping.

4.2 3D Depth Camera Core Source Code Analysis (`color_follow_2D.py`)

1. Depth Map Direct Extraction of True Distance

The depth camera subscribes to RGB and Depth through ApproximateTimeSynchronizer. After tracking the block centroid `(center_x, center_y)`, it directly indexes the true physical distance from the depth map matrix and calculates the horizontal geometric distance between vehicle and block:

```
# color_follow_2D.py -- True physical depth reading and horizontal distance derivation
cx_int = int(np.clip(center_x, 0, 639))
cy_int = int(np.clip(center_y, 0, 479))

cur_depth = depth_image[cy_int, cx_int] / 1000.0  # Convert mm to meters
pose_t = self.compute_heigh(center_x, center_y, cur_depth)
self.cur_distance = math.sqrt(pose_t[1]**2 + pose_t[0]**2) * 1000.0  # millimeters
```

2. 10-Frame High-Robustness Locking Strategy

For block reflection or slight jitter, the depth camera version uses a stricter 10-frame continuous center alignment mechanism (`ALIGN_COUNT_THRESHOLD = 10`). After confirming target stability, buzzer sounds and smoothly advances to 240mm:

```
# color_follow_2D.py -- State machine transition control
if is_aligned and self.Done_flag and not self.adjust_dist and not self.start_grasp:
    self.aligned_count += 1
    if self.aligned_count >= ALIGN_COUNT_THRESHOLD:  # Continuous stable 10 frames
        self.Beep_Loop()
        self.adjust_dist = True                      # Enter approach phase
```

```
        self.XY_Track_flag = False                    # Lock gimbal

if self.adjust_dist:
    if self.cur_distance > GRASP_DIST + 0.2:
        self.move_dist(self.cur_distance)
    else:
        if self.cur_distance > 200:
            self.adjust_dist = False
            self.start_grasp = True
            self.pubVel(0, 0, 0)
```

3. Gripper Adaptive Angle Computation and 3D IK Grasping

After chassis stops, the algorithm calculates joint 5 target angle based on block contour vertex slope, sends target 3D coordinates and rotation angle to `grasp_desktop`, and after superimposing 3D compensation from `offset_value.yaml`, descends for precise grasping:

```
# color_follow_2D.py —— Gripper adaptive alignment and message publishing
vx = int(self.corners[0][0]) - int(self.corners[1][0])
vy = int(self.corners[0][1]) - int(self.corners[1][1])
target_joint5 = compute_joint5(vx, vy)
self.joint5.data = int(target_joint5)

self.TargetJoint5_pub.publish(self.joint5)
self.pos_info_pub.publish(tag_msg)
```

Through this smooth adaptation of dual-camera solutions, whether using USB monocular lens or 3D depth camera, the robot can achieve smooth and fluid dynamic block servo tracking and autonomous grasping.

4.3 Grasping Offset Tuning (Both Camera Models)

If position deviation occurs during actual grasping (such as missing, knocking over objects, offset left/right, or hitting the table), you can directly modify the spatial offset configuration file for the corresponding camera:

1. Where to Modify

Configuration files are located in the `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/dofbot_pro_driver/config/` directory. Select based on camera type:

- **3D Depth Camera (Nuwa-HP60C)**: `offset_value.yaml`
- **USB Camera**: `usb_offset_value.yaml`

2. How to Modify

The file contains offset parameters in X, Y, Z directions (unit: meters, recommended adjustment per tweak is `0.005`~`0.01`, i.e., 5~10mm):

- `x_offset` **(forward/backward direction, positive toward vehicle front)**:
    - Gripper extends too far and knocks over object **decrease value** (adjust smaller or set negative)
    - Gripper doesn't extend far enough and misses **increase value** (adjust larger)

- `y_offset` **(left/right direction, positive toward vehicle left)**:
    - Gripper deviates right **increase value** (correct left)
    - Gripper deviates left **decrease value** (correct right)
- `z_offset` **(up/down height, positive toward ceiling)**:
    - Gripper too high in air and can't grasp **decrease value** (probe deeper)
    - Gripper hits or scrapes table **increase value** (raise higher)